

MEMS Resonant Accelerometer for Wearable Heartbeat Motion Detection

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Abstract—Ballistocardiography (BCG), the measurement of subtle chest vibrations generated during heart beats, requires a sensing architecture that is highly sensitive, low noise, and capable of rejecting environmental disturbances while remaining power efficient. BCG uses immovable structures such as couches and beds to read a patient’s heart information. This project proposes the design and modeling of the sensing modality of a wearable heart monitoring device. It incorporates a resonant accelerometer operating at nearly 10kHz and powered from a single 5V supply. The device integrates a double-ended tuning fork (DETF), a $399 \times 399 \mu\text{m}$ proof mass, c-shaped flexure and a double-sided drive and sense lateral comb drive actuator. Closed loop actuation is implemented using a $25 \text{ M}\Omega$ transimpedance amplifier (TIA), which is used to improve the parasitic feedthrough and provide a stable oscillation to the device. CoventorWare simulation was used to fine tune the resonance towards the 10 kHz target with a final design landing at 9.984 kHz in the first in-plane mode. Noise, parasitic capacitances, fabrication tolerance sensitivity, and quality factor performance were analyzed to determine the validity of the design. The finished design demonstrates a viable MEMS resonator ready for integration into a wearable BCG sensing device.

I. INTRODUCTION

Modern wearable monitoring devices have a demand for low power, miniaturized, high performance sensing nodes. BCG reads the subtle chest vibrations produced by the heart ejecting blood into the circulatory system and it provides various details about cardiac output such as the heart rate, stroke volume and blood pressure. Heart rhythms typically range from 5-30 μg on the chest wall which require specialized performance from the sensor. Conventional BCG setups place sensitive transducers under mattresses or beds, however, these setups are not suitable for continuous monitoring during daily life. An alternative is a wearable heart monitoring sensor that is placed on the chest. Capacitive accelerometers, which measure displacement directly, often compete with parasitic capacitances and amplifier noise while in resonant accelerometers,

external acceleration is translated into a frequency shift. This allows resonant accelerometers to filter out broadband noise and enable long, low power operation.

The goal of this project is to design, model, and implement a 10 kHz resonant accelerometer using the PolyMUMPs process and to demonstrate that it can be driven from a single 5 V battery. The proposed device integrates a $399 \mu\text{m} \times 399 \mu\text{m}$ proof mass supported by cantilever suspension beams, a DETF resonator for stiffness-dominated frequency setting, a rigid C flexure, and lateral comb electrodes for actuation and sensing. The closed-loop circuit uses transimpedance amplifier (TIA) that feeds motional current from the sense comb into the drive combs, creating a self oscillating system.

This report reviews the mechanical and electromechanical modeling, parasitic analysis, noise modeling, circuit architecture, and second-order performance considerations present in the proposed resonant accelerometer. These results demonstrate that the device satisfies the required specifications: 10 kHz nominal resonance, PolyMUMPs compatibility, and 5 V battery-driven operation.

II. THEORY AND BACKGROUND

A. Double-Ended Tuning Fork (DETF) Resonator

Several resonator types were considered, including clamped-clamped and free-clamped beams. Among the various resonator architectures, the double-ended tuning fork is the most rewarding due to its high stability and high quality factor [2]. The DETF moves in an anti-phase vibration which allows for external forces and noise to mathematically cancel out and removes common-mode forces. When force is applied through the proof mass, the effective stiffness and frequency changes. Tensile load on the DETF tines increase the stiffness while compressive load decreases the stiffness and frequency. The double ended tuning fork has two parallel beams with one

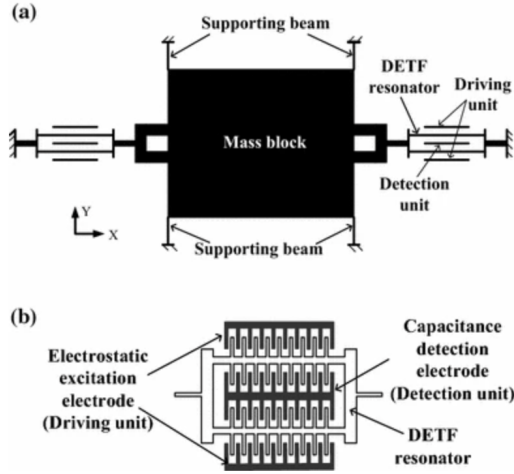


Fig. 1. Initial Sketch of Sensor [1]

side anchored while the other side is clamped to the c flexure. The DETF vibrates in an anti-phase mode where they move in opposite directions. A high frequency resonance is produced and it is very sensitive to stress induced by the electrodes. The resonant frequency of the DETF was measured with

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{k_{\text{DETF}}}{m_{\text{eff}}}}$$

where the k_{DETF} is the in-plane stiffness of the two beams and horizontal bar. m_{eff} is the effective mass. Inducing tension increases the stiffness and causes the frequency to increase while compressive stress reduces the stiffness and lowers the frequency. The DETF was chosen for this design due to its easiness with integrating a comb drive actuation and its high sensitivity to acceleration.

B. Electrostatic Actuation With Lateral Comb Drives

The sensor uses lateral comb drive actuation for in-plane oscillations. When the fixed and moving fingers are biased with different voltages, the overlapping surfaces of the comb create an electrostatic force.

$$F_x = \frac{1}{2} \frac{\partial C}{\partial x} V^2 = \frac{1}{2} \frac{N \epsilon_0 t}{g} V^2$$

where N = the number of drive fingers, t = the thickness of the fingers, g = the gap between the comb fingers and $V_{\text{eff}} = V_{\text{DC}} - V_{\text{drive}}$. The number(N) of drive finger pairs are 23 pairs per side of the proof mass and the thickness of the fingers (t) is equal to the thickness of POLY1. A double

sided acuation is used in this device to maintain symmetry and increase total force.

C. Motional Current and Capacitive Sensing

Motion of the DETF changes the overlap of the sense combs and produces a capacitance. The structure is biased at 5V while the sense fingers are at a virtual ground. The movement of the structure generates a motional current of

$$i_{\text{mot}}(t) = V_{\text{DC}} \frac{\partial C}{\partial x} \dot{x}(t)$$

where the number of sense combs on each DETF is 9 pairs. For the formula, $\frac{\partial C}{\partial x}$ is equal to 5.3×10^{-11} F/m

D. Electrical Equivalent Model of a Mechanical Resonator

The noise model of the resonator can be drawn as an equivalent RLC circuit where

$$L_x = \frac{m}{\eta^2}, \quad C_x = \frac{\eta^2}{k}, \quad R_x = \frac{\omega_0 m}{Q \eta^2}$$

is the electrostatic transconductance. $\eta = V_{\text{DC}} \frac{\partial C}{\partial x}$. This allows the resonator to be analyzed as part of an electrical feedback loop. The quality factor, which measures energy dissipation and influences stability, used in the device is 10000.

E. Noise Sources in Resonant Sensors

The prominent noise source that influence the minimum detectable signal is brownian mechanical noise. To find the brownian noise,

$$S_a^{1/2} = \sqrt{\frac{4k_B T \omega_0}{mQ}}$$

is used. The equation shows the noise produced by random movement of the particles in the system.

F. Parasitic Capacitances and Feedthrough

Parasitic capacitances will have an effect on the device performance. There are some more important capacitances such as the coupling between the drive and sense electrodes, C_{ODS} , and the electrode to substrate capacitances, C_{DB} and C_{SB} . In the closed loop actuation, it can create an electrical path that the sense electrodes might see as a signal even when the proof mass doesn't move. Because of the high quality factor and trans-resistance, this design is more sensitive to signals and will pick up C_{ODS} . This can reduce the apparent sensitivity, mess with the loop phase and make it harder to read small frequency changes. With the C_{DB} and C_{SB} , the motional current will become a smaller fraction. These

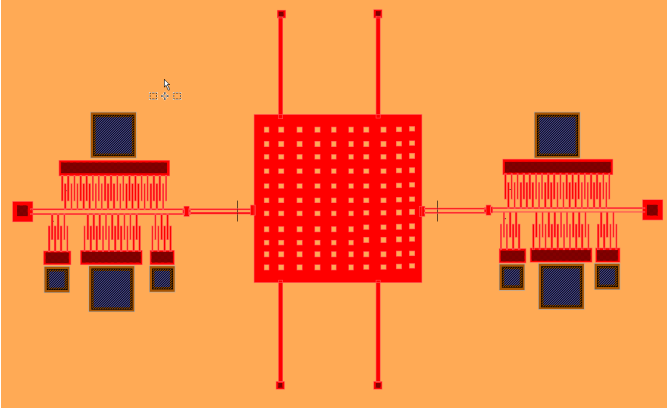


Fig. 2. Ledit Layout of Proposed Device

capacitances will contribute to the output capacitance from the microresonator and will be fed into the trans-resistance amplifier. The amplifier will see a higher capacitance due to the misleading parasitic capacitances and this will limit the bandwidth and increase the noise. Some less important parasitic capacitances are the electrode to ground, C_{01} , and resistive paths: R_{SUB} , R_{BACK} , R_{LEAK} , which are leakage paths that resistively couple the drive and sense electrodes due to moisture or impurities on the substrate. These resistive paths lead to drainage of the signal, bias drift and a reduction in the stability of the resonator's output.

G. Closed-Loop Oscillation Principle

To continue the oscillation near 10 kHz, the overall feedback loop must have a loop gain magnitude at resonance of 1. The loop is implemented by feeding the motional current to a transimpedance amplifier, which converts current to voltage, and the output is fed back to the drive combs. Because the sensing capacitance is very small due to limited sensing electrodes, a transimpedance gain of 25 M Ω is used to significantly amplify the current and make it viable for the drive electrodes.

III. DEVICE MODELING AND DESIGN

The proposed resonant accelerometer integrates a $399 \mu\text{m} \times 399 \mu\text{m}$ proof mass, a double-ended tuning fork (DETF) resonator, a C shaped flexure, and lateral comb electrodes for actuation and sensing. The process used is PolyMUMPs and the structure is made out of POLY1 with a $2 \mu\text{m}$ thickness. The overall architecture converts low frequency chest acceleration into a measurable frequency around 10 kHz.

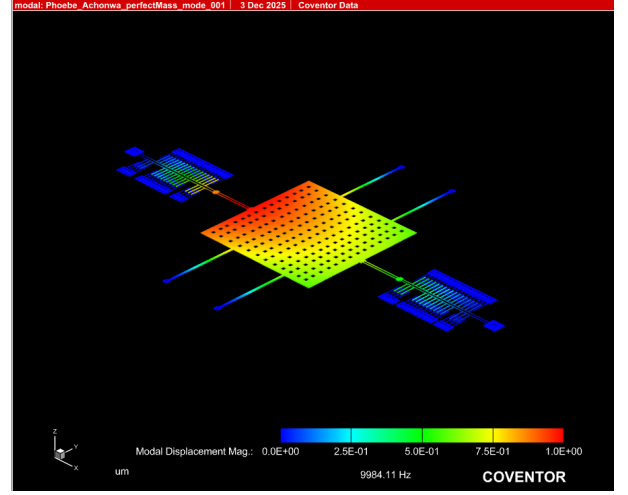


Fig. 3. CoventorWare Modal Analysis

A. Initial Design

The initial design iteration implemented a $200 \times 200 \mu\text{m}$ proof mass with a DETF driven using single-sided combs. Although mathematically accurate, the CoventorWare modal analysis showed that the architecture had some out-of-plane movement and a high resonant frequency. The few options to reduce the high resonant frequency was to decrease the resonator stiffness by decreasing the width and increasing the length of it tines, and the second option was to increase the proof mass by a considerable size. After several iterations of proof mass increasing and changing the dimensions of the detf tines, resonant frequency reached an in-plane mode of 9.984 kHz.

B. C Flexure Coupling Element

The c shaped flexure links the proof mass to the DETF and is used to transfer force while preventing unwanted out of plane motions. This structure was initially made rigid to keep the flexure from transferring any of its own movement. As iterations to reach the required resonant frequency showed a significant difference in the resonant frequency, the dimensions of the c shaped flexure began to grow to make the flexure softer. Although the flexure is softer, the stiffness is still much higher than the resonator stiffness (0.05-0.07 N/m).

C. Double-Ended Tuning Fork (DETF) Resonator

In the initial design, the DETF tines were $200 \mu\text{m}$ long and $10 \mu\text{m}$ wide, which set a resonant frequency higher than 10 kHz when combined with the relatively small proof mass. To bring the resonance down, the final design was changed

TABLE I
FINAL DESIGN PARAMETERS OF THE MEMS ACCELEROMETER

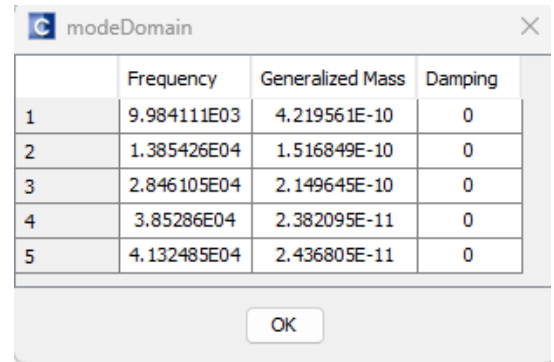
Subsystem	Parameter	Final Value / Specification
Proof Mass	Material	PolyMUMPs Poly1 (2 μm thick)
	Dimensions	399 $\mu\text{m} \times 399 \mu\text{m}$
	Estimated effective mass	$\approx 6.2 \times 10^{-11} \text{ kg}$
Suspension Beams	Count	4 beams (two per side)
	Length	300 μm
	Width	8 μm
	Thickness	2 μm
	Out-of-plane stiffness	$\approx 3\text{--}4 \text{ N/m}$
	(estimated)	
C Flexure	Leg length	120 μm
	Leg width	4 μm
	Gap between legs	4 μm
	Top bar span	12 μm
	Estimated stiffness	$\approx 6\text{--}8 \text{ N/m}$
Double-Ended Tuning Fork (DETF)	Tine length	300 μm
	Tine width	4 μm
	Tine spacing	4 μm
	Horizontal bar width	6 μm
	Effective stiffness	$\approx 0.05\text{--}0.07 \text{ N/m}$ (estimated)
	Verified in-plane resonance	9.984 kHz (Mode 1)
Drive Comb Actuators	Finger width	3 μm
	Gap	3 μm
	Overlap length	50 μm
	Finger count	23 moving + 23 fixed per side
	Bias voltage	5 V DC with AC drive
Sense Comb Electrodes	Finger count	9 moving + 9 fixed
	Static sense capacitance	$\approx 2.7 \text{ fF}$
	Capacitance sensitivity	$\approx 5.3 \times 10^{-11} \text{ F/m}$

to use 300 μm -long, 4 μm -wide tines, separated by a 4 μm gap and connected by a 6 μm -wide bar. The changes reduce the effective stiffness of the DETF and increase its sensitivity to axial stress. CoventorWare's modal analysis confirmed that with these dimensions and the updated proof mass, the first mode appears at 9.984 kHz, closely matching the 10 kHz target. Because the suspensions and C-flexure remain significantly stiffer than the DETF, this mode is clearly dominated by the tuning fork rather than by rigid-body motion of the mass.

D. Electrostatic Drive Electrode Design

Drive combs are placed on both sides of the DETF to implement double-sided actuation. Each comb finger is 3 μm wide with a 3 μm gap, 50 μm overlap length, and 2 μm thickness, following PolyMUMPs design rules. There are 23

moving and 23 fixed fingers on each side of the structure. The symmetric placement of combs significantly reduces torque and helps maintain an in-plane motion. This choice was made



	Frequency	Generalized Mass	Damping
1	9.984111E03	4.219561E-10	0
2	1.385426E04	1.516849E-10	0
3	2.846105E04	2.149645E-10	0
4	3.85286E04	2.382095E-11	0
5	4.132485E04	2.436805E-11	0

Fig. 4. Modal Analysis Frequencies

to reduce rotation and out of plane movement and provide a higher force.

E. Sense Electrode Design (9-Pair Comb)

The sense comb drive is implemented only on one side (in a D-S-D arrangement), with 9 moving and 9 fixed fingers using a $3 \mu\text{m} \times 3 \mu\text{m} \times 50 \mu\text{m}$ geometry. A larger sense array was initially used to boost motional capacitance, but in practice, there was no room to fit all the electrodes on the DETF. As increases to the DETF tines were made, more sensing electrodes could have been placed. The final 9-pair configuration provides a capacitance of 2.7 fF and a capacitance sensitivity of $\frac{\partial C}{\partial x} = 5.3 \times 10^{-11} \text{F/m}$, which works well when combined with the high feedback resistance and high Q-factor. The D-S-D layout was chosen to preserve the symmetric drive actuation while still dedicating a small region to sensing.

F. Parasitic Capacitance Estimation

Although there are parasitic capacitances that show up in the device, they don't alter the motional current because the sense electrodes are held at virtual ground by the TIA. Calculated values of the parasitic capacitances calculated are **Parasitic Capacitance Components**

Component	Value	Origin
C_{feed}	1–2 fF	Drive-sense overlap, D-S-D spacing
$C_{\text{sub,sense}}$	40–50 fF	Sense fingers to substrate + Poly0
$C_{\text{sub,drive}}$	60 fF	Drive fingers to substrate
C_{mass}	$\sim 690 \text{ fF}$	Proof mass to ground plane
Routing parasitics	5–20 fF	Lateral interconnect

G. Noise Modeling (Mechanical + Electrical)

The mechanical and electrical noise of the device can be modeled by finding the brownian noise and trans-impedance amplifier noise. Using the updated mass

$$m \approx 6.2 \times 10^{-11} \text{ kg}$$

, quality factor (Q) = 10000, and 9.984 kHz, the brownian noise comes out to $7.5 \times 10^{-6} \frac{\text{g}}{\sqrt{\text{Hz}}}$ while the TIA noise comes out to $0.64 \frac{\mu\text{V}}{\sqrt{\text{Hz}}}$.

The complete system utilizes a DETF spring mass damper model, electrostatic drive force input, motional current output and a TIA with 25 M Ω

IV. SECOND-ORDER EFFECTS

A. Fabrication & Temperature Tolerances

There are some lateral and vertical variations, that are common, on the order of $\pm 5\%$. These variations affect the stiffness and electrostatic transduction coefficients of the resonator which leads to a $\pm 5\%$ variation in the DETF stiffness which then corresponds to a resonance shift.

There can also be variations in the comb drive fingers which will change the electrostatic drive force and sense capacitance. These changes can come from variations in the fingers' gaps. The c flexure and suspension beams can also experience variations but because of their large stiffness compared to the DETF stiffness, the variations don't make a big contribution to the resonant changes. The resonance frequency may deviate by $\pm 3 - 5\%$.

B. Vacuum Level and Q Factor Sensitivity

The quality factor of the resonator is dependent on the vacuum level inside the package because air damping is the prominent reason for the energy loss. Under a medium vacuum environment, the 10000 quality factor can be achieved [3] [4]. If the vacuum degrades due to leakage, the squeeze film damping between the drive/sense fingers will grow and reduce the quality factor and attenuate amplitude. Quality factor variation effects the frequency, noise performance and loop gain.

C. Calibration and Tuning Approaches

To make up for the unexpected variations in fabrication, temperature, and vacuum level, a couple of strategies can be applied. With electrostatic spring-softening, a controlled DC bias can be applied to the drive combs to introduce a spring softening effect and corrects the frequency. Another method can be adding a small pair of tuning combs that are specifically meant for providing electrostatic stiffness without interfering with the original sensing signal. Adjusting the drive amplitude can stabilize oscillation and improve signal-to-noise ratio.

D. Major Energy Loss Mechanisms

Some mechanisms that contribute to energy loss in the resonator are anchor loss, and residual squeeze film damping. Anchor loss can occur as energy leak into the substrate through the DETF and suspension beams anchoring. Although vacuuming significantly minimizes air damping, it doesn't mean its effects are at 0. Altogether, these mechanisms can

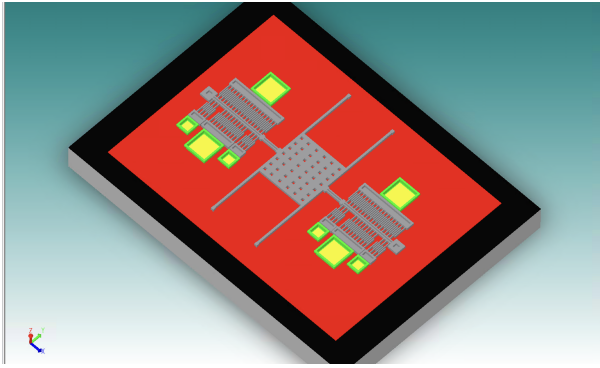


Fig. 5. 3D Model of Resonant Accelerometer

set a limit on the resonator's quality factor even with ideal packaging conditions.

E. Integrated Design Considerations

The final design integrates the electromechanical architecture and packaging assumptions into a cohesive system that satisfies the three primary project requirements: PolyMUMPs implementation, 10 kHz resonance, and 5 V operation. The resonant frequency is determined primarily by the DETF geometry and proof mass dimensions and is validated by CoventorWare modal analysis at 9.984 kHz for the first in-plane mode. The double-sided comb layout provides symmetric actuation, while the D-S-D pattern enables single-ended sensing. The 25 M Ω TIA ensures sufficient loop gain to overcome damping and maintain oscillation across likely variations in Q . Vacuum packaging and symmetric anchor placement help minimize energy loss and suppress out-of-plane motion. These elements form a self-oscillating resonant accelerometer architecture whose output frequency reflects the acceleration of the device, making it suitable as the sensing node of a wearable BCG sensor.

V. DISCUSSION

The final resonator design satisfies the project's main constraints: it is compatible with the PolyMUMPs process, resonates close to 10 kHz, and can be driven from 5 V. The DETF defines stiffness, the C flexure rigidly links to the proof mass, and the suspensions support the resonant frequency with minimal interference. Through multiple tunings of the proof mass and DETF dimensions, the in-plane mode was placed at 9.984 kHz with good separation from out-of-plane modes. Electrode geometry reflects realistic trade-offs where the nine-pair sense-comb has to be balanced between motional current and parasitic capacitance, while the

double-sided actuation was necessary to avoid torque-induced movement, maintain symmetry and provide a higher force. The design closely approaches the practical limits for PolyMUMPs and the footprint. Performance will be sensitive to vacuum level and to fabrication tolerances such as deviations in Q , DETF dimensions, or stress could push resonance outside target or prevent stable oscillation. Within the project scope, the design demonstrates that a PolyMUMPs-based resonant accelerometer is feasible as the core sensing element for a wearable ballistocardiography (BCG) system.

VI. CONCLUSION

A 10 kHz MEMS resonant accelerometer in PolyMUMPs is demonstrated, using a DETF resonator, a 399 $\mu\text{m} \times 399 \mu\text{m}$ proof mass, a C-flexure, and a double-sided lateral comb drive under a 5 V supply. Analytical and finite-element modeling confirmed a viable in-plane resonance at 9.984 kHz and showed that the design meets the project's principal requirements. A TIA-based readout circuit with a 25 M Ω feedback resistor was shown as sufficient for sustained self-oscillation. After evaluating second-order effects (fabrication tolerances, temperature, vacuum, damping) the design appears robust under realistic conditions — though sensitive to Q and geometry variation. Overall, the design proves that a compact, low-power, high- Q resonator can serve as the sensing core for a wearable BCG system. Future work can explore differential sensing, shielding, and integrated tuning to further improve stability and noise performance.

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